Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/01/2015
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Project Title:	Mechanisms for Plant Adaptation to Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/27/2012	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2018 per PI (Ed., 12/10/15) NOTE: Extended to 12/31/2015 per NSSC information (Ed., 12/16/14) NOTE: Extended to 12/31/2014 per NSSC information; original end date was 1/31/13 (Ed., 1/14/14) NOTE: Extended to 12/31/2013 per NSSC information; original end date was 1/31/13 (Ed., 1/25/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AN71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Successful plant growth in closed-loop life support conditions is a difficult challenge for the realization of long-term habitation of spacecraft and other extraterrestrial environments. In such environments, plants can undergo stress induced by a number of factors including changes in gravity, radiations, vibration, limited exchange of gases, and suboptimal growth conditions (temperature, light, nutrients). These sources of stress are often associated with reprogramming of gene expression and can cause limited plant growth, development, and yield. To facilitate plant life in space, it is crucial to acquire a better understanding of the genetic changes that enable plant cells to respond to spaceflight stress. To do so, one goal of this proposal is to define the underlying mechanisms of plant adaptation to the spaceflight environment at a transcriptional level. Through research supported by NASA, we have successfully identified a protein named AtIRE1 as a master regulator of transcription in conditions of stress responses to abiotic, biotic stress and to gravity changes in plants. However, the downstream signaling pathways controlled by AtIRE1 are not well defined, especially in conditions of altered gravity. A better understanding of the signaling pathways controlled by AtIRE1 is important to understand how plants can grow in conditions of stress. We propose to use in-flight and ground resources along with genomics and transcriptomics analyses in the model plant Arabidopsis thaliana to understand the regulatory role of AtIRE1 in adaptation to spaceflight stress. Development of our research will continue to contribute to the understanding of basic signaling pathways that are in place to ensure stress survival in hostile environments, thus making possible the design and growth of plants that are resistant to space stress. To contribute further to the successful realization of habitation in space, we also aim to develop plants that can function as bioindicators of stress during in-flight situations. To do so, we will engineer plants with an AtIre1 substrate that is activated specifically in conditions of stress and that we will adapt to function as a visual stress reporter. The availability of real-time stress bioindicators will provide scientists and astronauts with direct read-outs of stress in the space environment. The results gathered in our research will contribute to NASA's strategic plans for the realization of long-term habitation of space and planetary surfaces. Because of the conservation of stress sensing and response mechanisms across multicellular organisms, we expect that our results will also have important implications for the general knowledge of stress and in the design of solutions for space stress management in multicellular organisms, including humans.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Plants are the primary source of energy and oxygen for all biospheric development. Thus, plants are essential to the establishment of large-scale controlled ecosystems on terrestrial and extraterrestrial surfaces. Similar to animals, plants suffer from stress linked to a variety of challenges. For example, certain adverse environmental conditions, such changes in temperature, reduced nutrients, and disease, as well as physiological situations, such as growth, can generate an imbalance between demand and capacity of protein synthesis in the endoplasmic reticulum (ER), a condition that is also known as ER stress (1, 2). The ER is the organelle that initiates the production of all secretory proteins. Secretory proteins constitute one third of the cellular proteome and allow cells to communicate with the external environment. The ER can sense stress due to accumulation of unfolded proteins in its lumen, and restores homeostasis by invoking a largely conserved cytoprotective signaling pathway, called the unfolded protein response (UPR). Compromised UPR leads to serious conditions and even death in animals and plants (3-5).
In animals and plants, several sensors and transducers initiate the UPR; however, IRE1, an ER-associated protein kinase and ribonuclease, functions as a major ER stress sensor and plays a key role in the UPR signaling by controlling UPR gene expression (4, 6, 7). Activities in our lab and others have demonstrated that plant IRE1 is needed to respond to a diversified portfolio of stimuli linked to abiotic and biotic stress and cell growth and differentiation, as well as to modulate responsive outputs (4, 5). Based on the key role of the Arabidopsis IRE1 (AtIRE1) in stress responses, our work aims to identify the genes that are under the transcriptional control of AtIRE1 in plants during growth as well as under stress due to enhanced secretory protein accumulation.

In all eukaryotes, IRE1 activates the UPR by an unconventional splicing of a bZIP transcription factor, namely Hac1 in yeast8-10 and Xbp1 in metazoans (11, 12). Recent efforts in our lab and others have shown that in the model species Arabidopsis thaliana, the functional equivalent of Hac1 and Xbp1 is bZIP60 (5, 13, 14). The bZIP60 mRNA splicing is followed by the synthesis of the mature form of bZIP60, which can modulate transcription of UPR genes in the nucleus. Thus, synthesis of the transcriptionally active bZIP60 occurs only in conditions of stress. In our research we will adopt the stress-dependent activation of bZIP60 to generate plants that can function as real-time stress bioreporters that can be easily used to report on the onset of stress in plants.

We believe that we will learn the mechanisms that enable plants to respond to ER stress. This is highly significant for the understanding of basic defense mechanisms in plants, and for engineering plants that can cope with stress on Earth.

We also expect that our research will have an important impact in biology with implications well beyond plants. Because the signaling pathways between plant and mammalian UPR are largely conserved, it is likely that understanding the UPR in a tractable multicellular model system, such as Arabidopsis, will allow building approaches with a strong potential to contribute to the solution of the problems in humans confronting cellular stress.

We also expect that we will be able to develop transplantable bioindicators of stress without the need of sophisticated instrumentation. These reporters will potentially facilitate the optimization of plant cultivation in hostile environments, such as marginal lands. In addition, because of the conservation of UPR mechanisms in plants and humans, real-time stress bioindicators will likely reflect the perception of stress in other organisms besides plants and thus they may be useful to develop practices to improve quality of life for humans.

In conclusion, not only will our research improve our nation's scientific competitiveness, it will also benefit education and the quality of life on Earth. Life on Earth depends on plants. With sharply growing environmental concerns (global warming, pollution, necessity to expand agriculture to marginal lands, etc.), understanding how plants can resist stress in hostile environments and adopting bioindicators help research to expand agriculture practices and thus ensure continued survival of the ever-growing human population.

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Task Progress:

Through the BRIC-18-2 (Biological Research in Canisters) experiments, we were able to fly wild type and a UPR mutant to the International Space Station (ISS), and carry out comparative transcriptomics analyses ISS-seedlings exposed to space conditions for 7 and 14 days. Seedlings of identical genotypes were also grown on ground, as control. The experiment consisted of two biological replicates, except for 14-day-old wild type that was based on only 1 biological replicate to make space for HOBO temperature data logger. Seeds were kept in dormant state through cold stowage. Upon docking of the SpaceX-3 Dragon capsule at the ISS, the experiment proceeded in the BRIC-18-2 at room temperature, which allowed seed germination. On-orbit canisters containing the seedlings were actuated by RNALater injection after 7 and 14 days, respectively from docking. This procedure was followed by a three-hour storage at room temperature, and then the canisters were stored at +4°C prior to longer term storage in the Minus Eighty-degree Laboratory Freezer until de-integration from the BRIC hardware and subsequent turnover to our team. SpaceX-3 Dragon containing our experiment launched on 4/18/14 and the experiment returned on ground 05/17/2014. In parallel, the experiment was replicated on ground at the Kennedy Space Center Test and Operations Support Contract Center (KSC-TOSC) in the International Space Station Environment Simulator (ISSES) Chamber.

All flight and ground samples were returned to us without contamination and we observed germination rates of at least 90%. Etiolated hypocotyl growth looked great. Both flight and ground control RNA samples were inspected for RNA quality and sequenced through customary approaches. Transcript reads were aligned to matching locations in the reference genome using dedicated software and transcript abundance was also estimated. The results that we obtained are statistically significant because of the number of technical replicates. We are now confirming the expression of some marker genes by quantitative RT-PCR to add robustness to our data for publication.

Our experiments to date have enabled us to establish differential gene regulation in wild type but also in mutant Arabidopsis with defects in pro-survival management strategies. These findings open new areas of research on plant strategies on stress management in space. We are now confirming transcriptomics results for specific genes to validate the RNASeq analyses and submit the results for publication.

Bibliography Type: Description: (Last Updated: 12/08/2025)

Articles in Peer-reviewed Journals Brandizzi F, Frigerio L, Howell SH, Schäfer P. "Endoplasmic reticulum-shape and function in stress translation." Front Plant Sci. 2014 Sep 1;5:425. eCollection 2014. http://dx.doi.org/10.3389/fpls.2014.00425 ; PubMed PMID: 25225498 , Sep-2014

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